Olist E-Commerce Business Intelligence & Performance Analytics Project Report

Project Objective

To analyse data from Olist's Brazilian e-commerce marketplace through data cleaning and validation, data modelling, interactive visualisation in Tableau, and SQL-based analysis in Google BigQuery to evaluate business performance and derive actionable insights.

Scope

Item	Detail
Period	2016 to 2018
Orders analysed	99,441
Currency	Brazilian real (R$)
Tools	Microsoft Excel, Power Query, Tableau, Google BigQuery

Data Source:

Brazilian E-Commerce Public Dataset by Olist (Kaggle),
https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

This dataset has 9 files, but this project analyses 8 files and excludes the geolocation file because the analysis is at the state level.

Methodology

Data cleaning (Excel and Power Query)

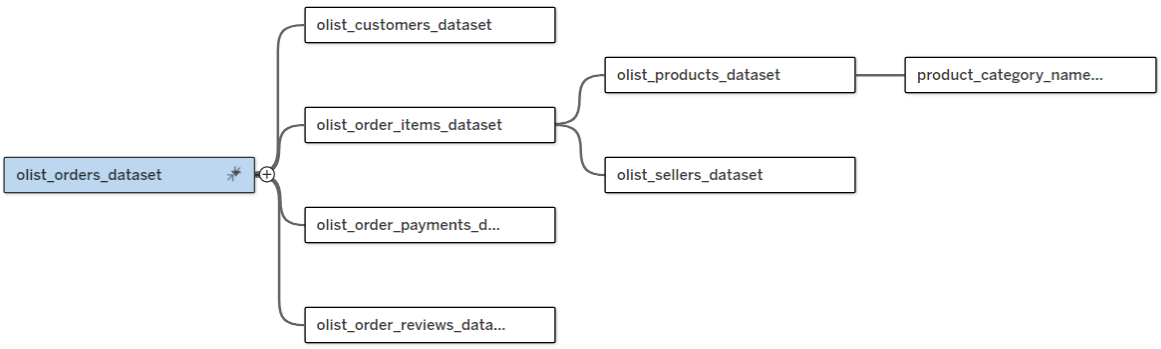
Cleaned each table separately and validated it against the original file.

1. Promoted headers and set the correct data type for every column.
2. Checked duplicates on each table's key (none were found).
3. Handled missing values: 610 missing categories filled as "unknown"; missing delivery dates kept blank because those orders were not delivered.
4. Validated row counts and totals against the source.

Cleaning log

Table	Issue found	Rows	Action
orders	Dates stored as text	All	Converted to Date/Time
orders	Delivered but no delivery date	8	Kept, noted
orders	Canceled but has a delivery date	6	Kept, noted
orders	Missing approval/carrier/delivery date	160 / 1,783 / 2,965	Kept blank: not yet approved, shipped, or delivered
customers	customer_id is new for every order	99,441 vs 96,096	Count customers with customer_unique_id
order_items	order_item_id is a line number, not a quantity	n/a	Count rows for items sold (112,650)
payments	Payment value of 0	9	Kept (6 vouchers, 3 not_defined)
reviews	Same review_id on several orders	789 IDs (1,603 rows)	Kept: valid
reviews	Review_comment_title and review_comment_message	All	Columns removed
products	Missing category	610	Filled with "unknown"
product_translation	2 categories have no English name.	2	Handled in Tableau with IFNULL().
products	Misspelled column names ("lenght")	2 columns	Renamed to "length"

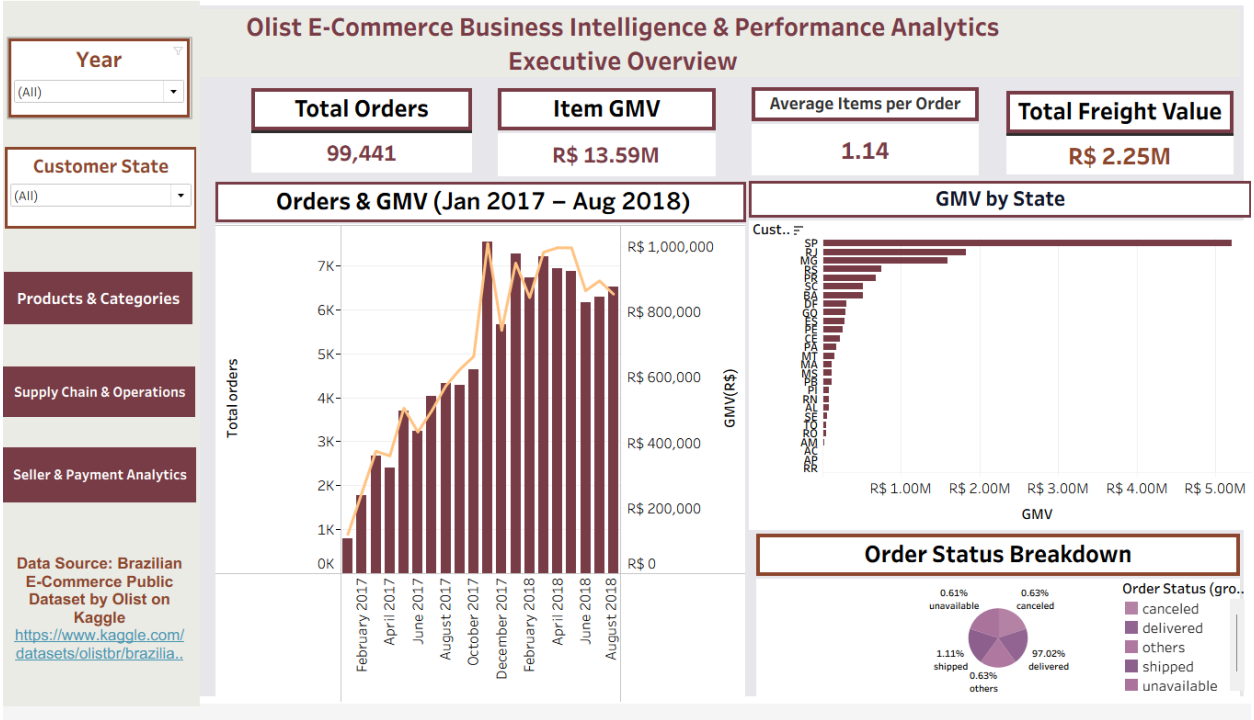
Data modelling (Tableau)



The 8 cleaned tables were linked with Tableau relationships.

Dashboards (Tableau)

Dashboard 1 – Executive Overview

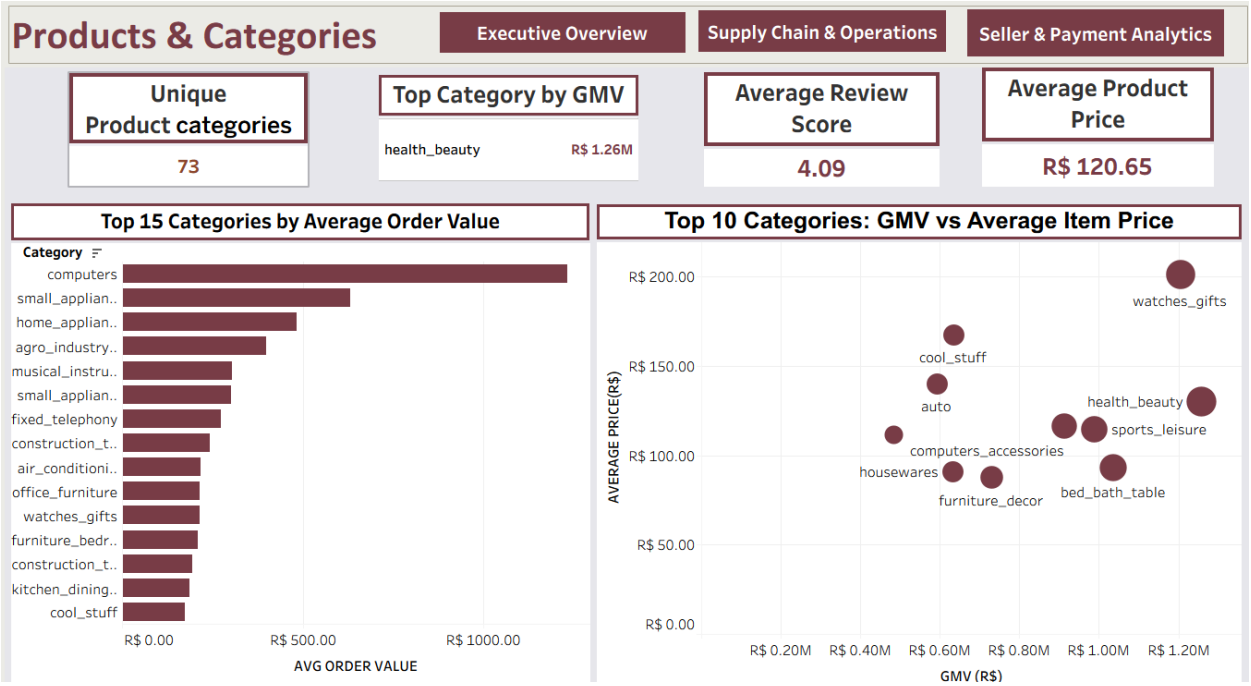


KPI	Value
Total orders	99,441
Item GMV	R$ 13.59M
Average items per order	1.14
Total Freight Value	R$ 2.25M

Insights

- Scale**:** Olist handled 99,441 orders worth R$ 13.59M in item GMV, with R$ 2.25M charged as freight.
- Growth: Monthly orders grew from 800 in January 2017 to 7,544 in November 2017. November 2017 was the peak sales month. After that, growth stabilised through 2018.
- Sales are concentrated in some regions: SP brings R$ 5.20M GMV, and SP, RJ, and MG together bring R$ 8.61M GMV.
- Freight cost: Freight equals about 16.6% of item GMV.
- Order Fulfilment: 97.02% of orders were delivered.

Dashboard 2 – Products & Categories

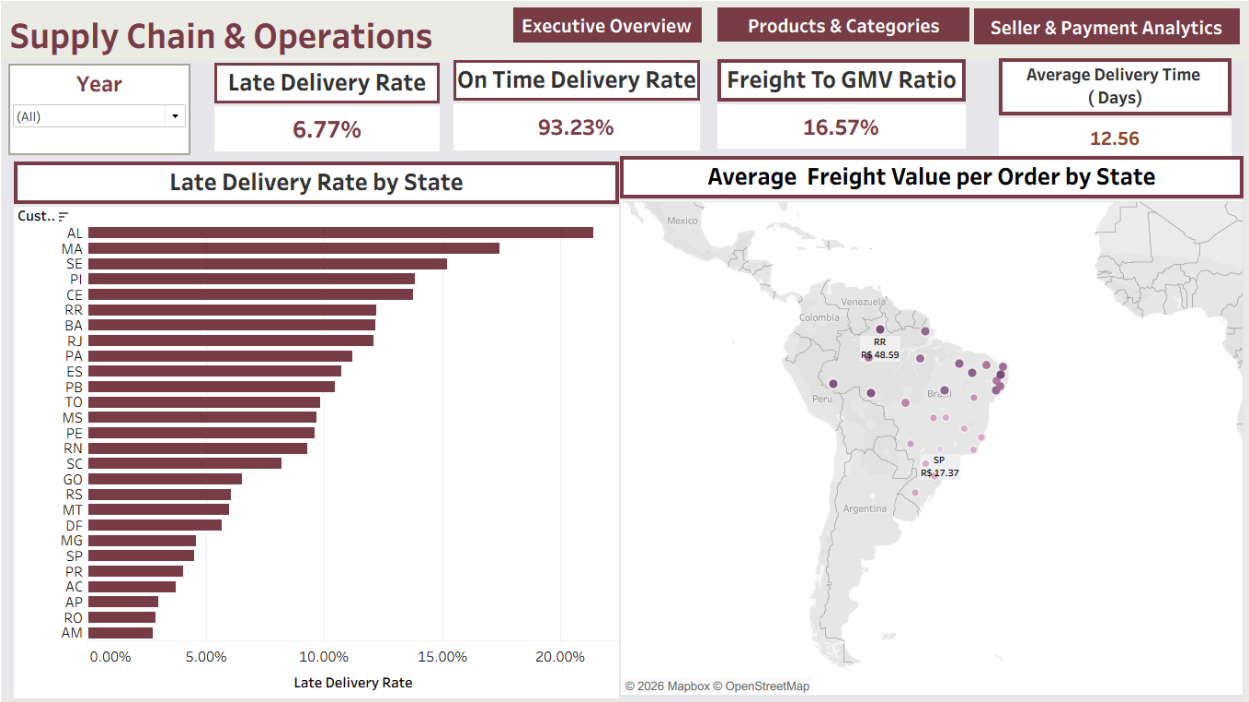


KPI	Value
Unique Product Categories	73
Top Category by GMV	health_beauty (R$ 1.26M)
Average Review Score	4.09
Average Product Price	R$ 120.65

Insights

- Product range: There are 73 product categories (excluding "unknown") with an average product price of R$ 120.65.
- Top categories by GMV: health_beauty (about R$ 1.26M) and watches_gifts (about R$ 1.21M) lead, followed by bed_bath_table and sports_leisure.
- Highest order value: computers have by far the highest average order value (R$ 1231.84).
- Volume vs price**:** bed_bath_table has the third-highest GMV (R$ 1.04M) at an average item price of about R$ 93.30, below the overall average of R$ 120.65.

Dashboard 3- Supply Chain & Operations

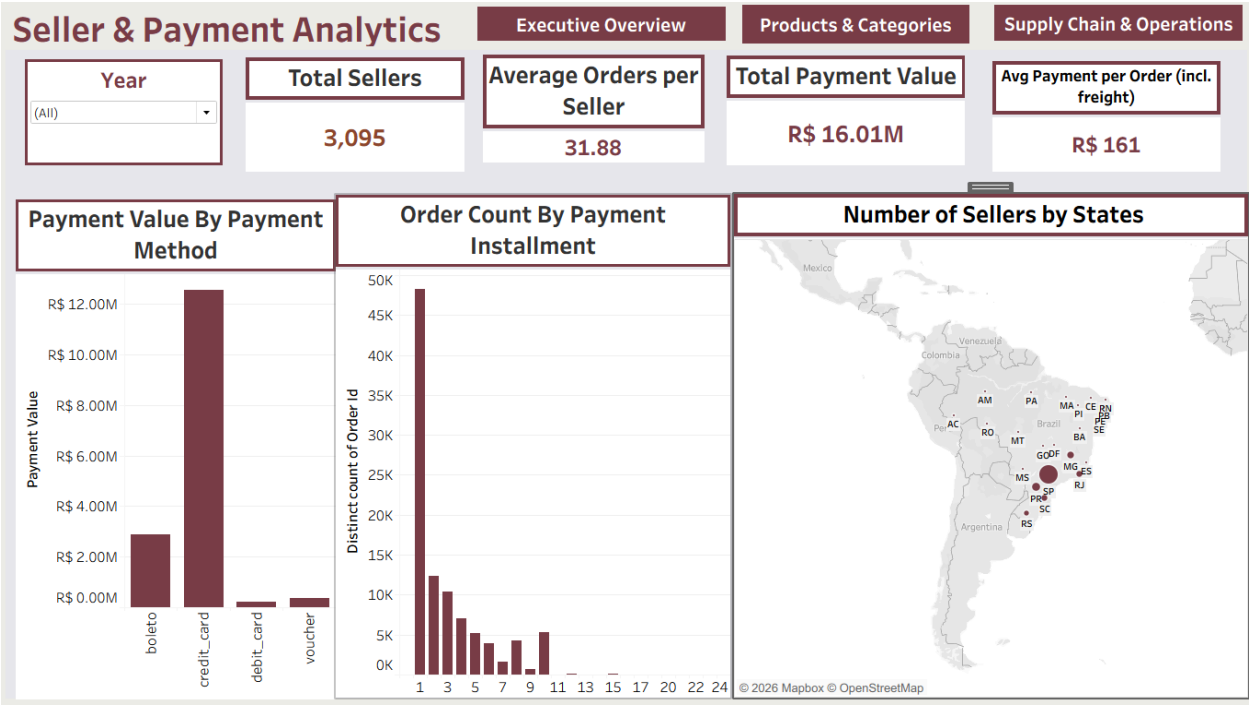


KPI	Value
Late delivery rate	6.77%
On-time delivery rate	93.23%
Freight to GMV ratio	16.57%
Average delivery time	12.56 days

Insights

- 93.23% of orders arrive on time and 6.77% late, with an average delivery time of 12.56 days.
- Freight cost: Freight value is 16.57% of GMV.
- Some states lag: Late delivery is highest in AL (21.41%), MA (17.43%), and SE (15.22%).
- Freight varies by region: Average Freight per order is highest in customer state RR (R$ 48.59) and lowest in SP (R$ 17.37).

Dashboard 4- Seller & Payment Analytics



KPI	Value
Total sellers	3,095
Average orders per seller	31.88
Total payment value	R$ 16.01M
Average payment per order	R$ 161

Insights

- Sellers: There are 3,095 sellers, averaging 31.88 orders each.
- Payment value: Total payment value is R$ 16.01M, averaging R$ 161 per order.
- Credit card dominates: Credit card carries about R$ 12.54M of payment value. Boleto is next at about R$ 2.87M, debit card at R$ 0.22M, and voucher at R$ 0.38M.
- Instalments: About 48,268 orders are paid in one instalment.

- Sellers are concentrated; SP has the most sellers (1849).

SQL analysis (Google BigQuery)

The cleaned tables were uploaded to BigQuery, and queries answered questions using SELECT, WHERE, GROUP BY, ORDER BY, LIMIT, JOIN, CASE, COUNT, COUNT DISTINCT, AVG, MIN, MAX, ROUND, EXTRACT, DATE, DATE_DIFF.

Q1	How many orders per year?	EXTRACT, COUNT, GROUP BY, ORDER BY	**year orders** **2016 329** **2017 45101** **2018 54011**
Q2	Is shipping to another state slower and costlier?	JOIN, CASE, AVG, DATE_DIFF, GROUP BY	**route items avg_freight avg_delivery_days** **Different state 70328 23.63 15** **Same state 39861 13.45 7.9**
Q3	How many instalments do payments use?	COUNT, GROUP BY, ORDER BY, LIMIT	**payment_installments payments** **1 52546** **2 12413** **3 10461** **4 7098** **10 5328** **5 5239** **8 4268** **6 3920** **7 1626** **9 644**

	Q4	What are the cheapest and most expensive items?	MIN, MAX, AVG, ROUND	Cheapest most_expensive average_price 0.85 6735 120.65
	Q5	What is the average payment by payment method?	COUNT, AVG, ROUND, GROUP BY, ORDER BY	payment_type payments avg_payment credit_card 76795 163.32 boleto 19784 145.03 debit_card 1529 142.57 voucher 5775 65.7 not_defined 3 0
	Q6	Which categories list the most products?	COUNT, GROUP BY, ORDER BY, LIMIT	**product_category_name products** **cama_mesa_banho 3029** **esporte_lazer 2867** **moveis_decoracao 2657** **beleza_saude 2444** **utilidades_domesticas 2335** **automotivo 1900** **informatica_acessorios 1639** **brinquedos 1411** **relogios_presentes 1329** **telefonia 1134**
	Q7.	Which cities have the most customers? (Top 10)	COUNT DISTINCT, GROUP BY, ORDER BY, LIMIT	**customer_city customers** **sao paulo 14984** **rio de janeiro 6620** **belo horizonte 2672** **brasilia 2069** **curitiba 1465** **campinas 1398** **porto alegre 1326** **salvador 1209** **guarulhos 1153** **sao bernardo do campo 908**

Insights

- Cross-state orders take longer and cost more: 15.0 days vs. 7.9, and freight of R$23.63 vs. R$13.45.
- Customers are concentrated in a few cities; the top 10 cities have 33804.
- Credit Card has the highest average payment (R$ 163.32) against R$ 65.70 for voucher.

Recommendations

Priority	Action	Evidence	KPI to track
1	Review delivery estimates and carriers in the states with the most late orders.	Late delivery is highest in AL (21.41%), MA (17.43%), and SE (15.22%).	Late delivery rate by state, freight to GMV ratio
2	Reduce dependence on one region by growing sales and sellers in other regions.	SP brings R$ 5.20M GMV, and SP, RJ, and MG together bring R$ 8.61 M GMV.	Share of GMV by state, number of sellers by state.
3	Keep credit card and instalment payment options available.	Credit card carries about R$ 12.54 M of payment value, against about R$ 2.87M for boleto and 48,268 orders are paid in one instalment.	Payment value by method, order count by instalment.
4	Focus category promotion and stock on the top-selling and high-value categories.	health_beauty (R$ 1.26M) and watches_gifts (R$ 1.21M) lead GMV. Computers have the highest average order value (R$ 1231.84).	Item GMV by category, average order value

Conclusion

Olist has a large marketplace: 99,441 orders, R$ 13.59M in item GMV, a 4.09 average review score, and 93.23% of orders delivered on time. Sales and sellers are concentrated in SP, late deliveries are highest in AL, MA, and SE, and freight is highest in RR. The four dashboards give a single view to track these measures.

Limitations

- Months covered**:** Orders run from September 2016 to October 2018, but September and October 2018 hold only 20 orders, so 2018 is effectively January to August.
- Costs and margins are not in the dataset, so findings are about revenue, not profit.
- 610 products have no category and appear as "unknown".

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Project files and links

Item	Location
Dataset	Kaggle – Brazilian E-Commerce Public Dataset by Olist
Dashboards	https://public.tableau.com/views/OlistE-CommerceBusinessIntelligencePerformanceAnalytics/ExecutiveOverview
GitHub Repository link	https://github.com/navitaarora/olist-ecommerce-analysis-